Punam Adhikari

AI Engineer | Data Science

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AI Engineer and M.Sc. Data Science student at TU Dortmund University, with experience in **speech AI, LLM fine-tuning, multimodal RAG, and data engineering**. Builds machine learning systems from model experimentation and evaluation through APIs, data pipelines, and production deployment.

PROFESSIONAL EXPERIENCE

AI Engineer · NSDevil — Oct 2025 – Present
Lalitpur, Nepal | On-site

- Fine-tuned **Whisper, NeMo, and Conformer-based ASR models** on Korean speech datasets, achieving **under 10% word error rate**.
- Developed **Korean text-to-speech systems with ESPnet**, integrating model inference and evaluation into the speech-AI workflow.
- Built a **microservice-based Korean essay-scoring system** using rubric-driven LLM evaluation to deliver structured, explainable feedback.
- Fine-tuned **HyperCLOVAX 3B and 14B** models using supervised fine-tuning, DPO, LoRA, and QLoRA across academic domains.
- Designed and deployed **multimodal RAG systems and Transformer models** on remote GPU infrastructure with CI/CD pipelines.

Software Engineer I · Abacus Insights — Jun 2025 – Oct 2025
On-site

- Designed and maintained **healthcare ETL and data pipelines** using Databricks, Snowflake, Python, PySpark, and SQL.
- Applied **NLP-based PDF extraction** to improve information retrieval from documents in the US healthcare domain.
- Monitored and troubleshot pipelines, optimized data quality and performance, and translated stakeholder requirements into scalable data solutions in an Agile environment.

AI/ML Research Intern · Danson Solutions — Nov 2024 – Apr 2025
Hybrid

- Engineered a **PHP-to-Java code translation pipeline** using Meta-LLaMA and GPT-3.5 Turbo to automate code migration.
- Fine-tuned and evaluated LLMs for code generation; explored AI-assisted refactoring with embeddings and syntactic analysis, and documented research findings.

EDUCATION

M.Sc. Data Science · TU Dortmund University — Oct 2026 – Present
Dortmund, Germany

Bachelor's in Computer Engineering · Pulchowk Engineering Campus — May 2021 – Jun 2025
Kathmandu, Nepal

Higher Secondary (+2 Science) · Satyawati Secondary School — 2018 – 2020
Tanahun, Nepal

SKILLS

Languages	Python, SQL, Java, Unix/Linux and shell scripting
Frameworks & Libs	PyTorch, TensorFlow, Keras, Scikit-learn, FastAPI, NumPy, Pandas
Data Engineering	Apache Spark, PySpark, Airflow, Databricks, Snowflake, Kafka, Azure Data Factory
Tools & Platforms	Git, Docker, CI/CD, AWS S3, Azure, Power BI, Tableau
Databases	PostgreSQL, MySQL, Oracle, MongoDB
Data & AI	LLM fine-tuning, RAG, NLP, speech AI, computer vision, model evaluation, statistical analysis

PROJECTS

CiteMind-AI-RAGOps · Retrieval & AI Operations
Python, LangGraph, Qdrant, FastAPI, React, Docker, Kubernetes, Prometheus
Built a research assistant combining **hybrid retrieval, cross-encoder reranking, and source-grounded responses**. Added containerized services, Kubernetes deployment configuration, and Prometheus monitoring to make retrieval and operations inspectable.

DevPilot · Agentic AI & Developer Tools
Python, MCP, FastAPI, Docker, OpenTelemetry
Developed a **human-guided debugging workbench** that investigates repository failures with MCP tools, proposes approval-gated edits, and verifies repairs through isolated tests with auditable artifacts.

VerityLake · Data Engineering & Governed RAG
Airflow, DuckDB, Delta Lake, MinIO, Chroma, Ollama, FastAPI
Engineered a data-to-RAG pipeline connecting ingestion, validation, versioned data, and vector indexes. Implemented **quality-gated publication, traceable citations, and reversible releases** to preserve data lineage through retrieval.

LLM120 · Language-Model Training Infrastructure
PyTorch, Transformers, Byte-Level BPE, BF16, SFT, DPO, RLOO
Implemented a **from-scratch 125M-parameter language-model training pipeline** with tokenizer training, mixed precision, and resumable checkpoints. Designed separate supervised fine-tuning, preference-optimization, and reward-model/RLOO stages.

LinguaVoice · Multilingual Speech
Whisper, XTTS-v2, MMS, Docker
Built a **five-language speech-to-text and text-to-speech platform** with browser recording, voice controls, training pipelines, and word/character error-rate evaluation.

AI Online Proctoring · Real-Time Computer Vision
Python, OpenCV, YOLO, Dlib, Scikit-learn, CNN
Built a **seven-module exam-monitoring pipeline** covering person/object detection, face verification, spoof detection, gaze, mouth movement, and head pose, with live alerts and lower-end hardware optimization.

FELLOWSHIPS & TRAINING

Computer Vision & Deep Learning Summer Internship · IGDTUW
Selected to represent Pulchowk Campus (IOE) in an international academic internship focused on computer vision, deep learning, and collaborative research.

AI Fellowship · Fuse Machines
Selected for a competitive fellowship recognizing technical excellence and potential in artificial intelligence.

CERTIFICATIONS

Machine Learning Specialization · Andrew Ng / Coursera
IBM Data Engineering Specialization · IBM
AWS Academy Data Engineering Certificate · AWS Academy
SQL Certificates – Intermediate & Advanced · HackerRank
Hugging Face Agents Course · Hugging Face
Model Context Protocol: Advanced Topics · Anthropic

VOLUNTEERING & COMMUNITY ENGAGEMENT

- Volunteered in **Nepal Red Cross blood donation programmes**, supporting community-based humanitarian service.
- Contributed to the delivery of **cultural events** through volunteer support and community participation.
- Hosted a **Nepal Scouts conference**, helping deliver the conference programme.

LANGUAGES

Nepali – Native **English** – C1
German – B1; actively learning and committed to reaching professional fluency.